

Description of the tracing file `blnc_SPb.trc`

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This file belongs to the suite of tracing files produced by SOLPS-ITER transport code. They were originally developed by Andrei Kukushkin in order to trace the time history of neutrals-related quantities computed by Eirene. Due to presence of a Monte-Carlo noise, such a tracing seems to be an essential tool to estimate and visualize the time-averaged (i.e. smoothed on a time scale much larger than the time step) evolution of the code output quantities.

The file `b2blnc_SPb.trc` is devoted to the particle balance, and all the traced quantities have the dimension of a number of particles crossing the certain boundary per second, i.e. s^{-1} . This file is a successor of original Kukushkin's `b2blnc.trc`, however, some quantities are removed (to reduce the amount of stored data) or revised to make the output consistent with the up-to-date physical model with drifts. The original file `b2blnc.trc` remains unchanged.

This document describes the content of the file `b2blnc_SPb.trc` after the revision and the adaptation for runs with drift turned on either in standalone or coupled mode of SOLPS-ITER code runs. This revision is made at 04 June 2018 by Ilya Senichenkov.

The first line of the file `blnc.trc` looks like

```
# 20180622 4 16 # # b2mn B2.5 bjb99/7 SOLPS-ITER 2018-06-23 15:56 ....
```

Here the 20180622 is the date of the last modification (hard-coded in the FORTRAN source file `$$$SOLPSTOP/modules/B2.5/srs/user/b2blns.F`, then it is typed the number of B2 isonuclear sequences (in this example it is 4 for D, He, Ne and Be), and then — the number of traced quantities for each of these sequences (in the version considered it is 16). After the double `# #` sign the technical information on code version, user, date of compilation, etc is provided.

The second line of `blnc_SPb.trc` contains the headers of traced quantities started by the `time` as zeroth column `# time ion_core_D ntr_core_D ...`

The full list of headers is

1. `ion_core`
2. `ntr_core`
3. `ion_targ`
4. `ntr_targ_tot`
5. `ntr_targ_rec`
6. `ntr_targ_spt`
7. `ntr_targ_pmp`
8. `ion_wall`
9. `ntr_wall_tot`
10. `ntr_wall_rec`
11. `ntr_wall_spt`
12. `ntr_wall_pmp`
13. `scr_ioniz`
14. `ntr_puff`
15. `flux_tot`
16. `ion_dn/dt`

Before starting the detailed description it is worth making two remarks

i) Most of the quantities are particle fluxes through one of the computational domain boundary segment. The sign convention used here is that the flux is positive if it is directed inside the domain and negative if it is directed outside. Note that this sign convention differs from ones used either in B2 or in Eirene.

ii) Since the computational domains are different in standalone and coupled modes, the meanings of traced quantities also depend on the mode. These two different modes are described separately.

Figure 1 illustrates the computational domain and its boundaries to make further references easier.

1 Coupled mode

1.1 Ion flux through the core boundary, `ion_core`

This is simply the integral of the ion flux density over the core boundary, summing over all ionized states corresponding to given isonuclear sequence is assumed. The integral is computed basing on the `fna` array for the versions 5.0 and earlier, and basing on the `fna.mdf` array for versions 5.2 and later. Which of two arrays is used is controlled by the switch `btfnb.mdf`.

Expected to be positive.

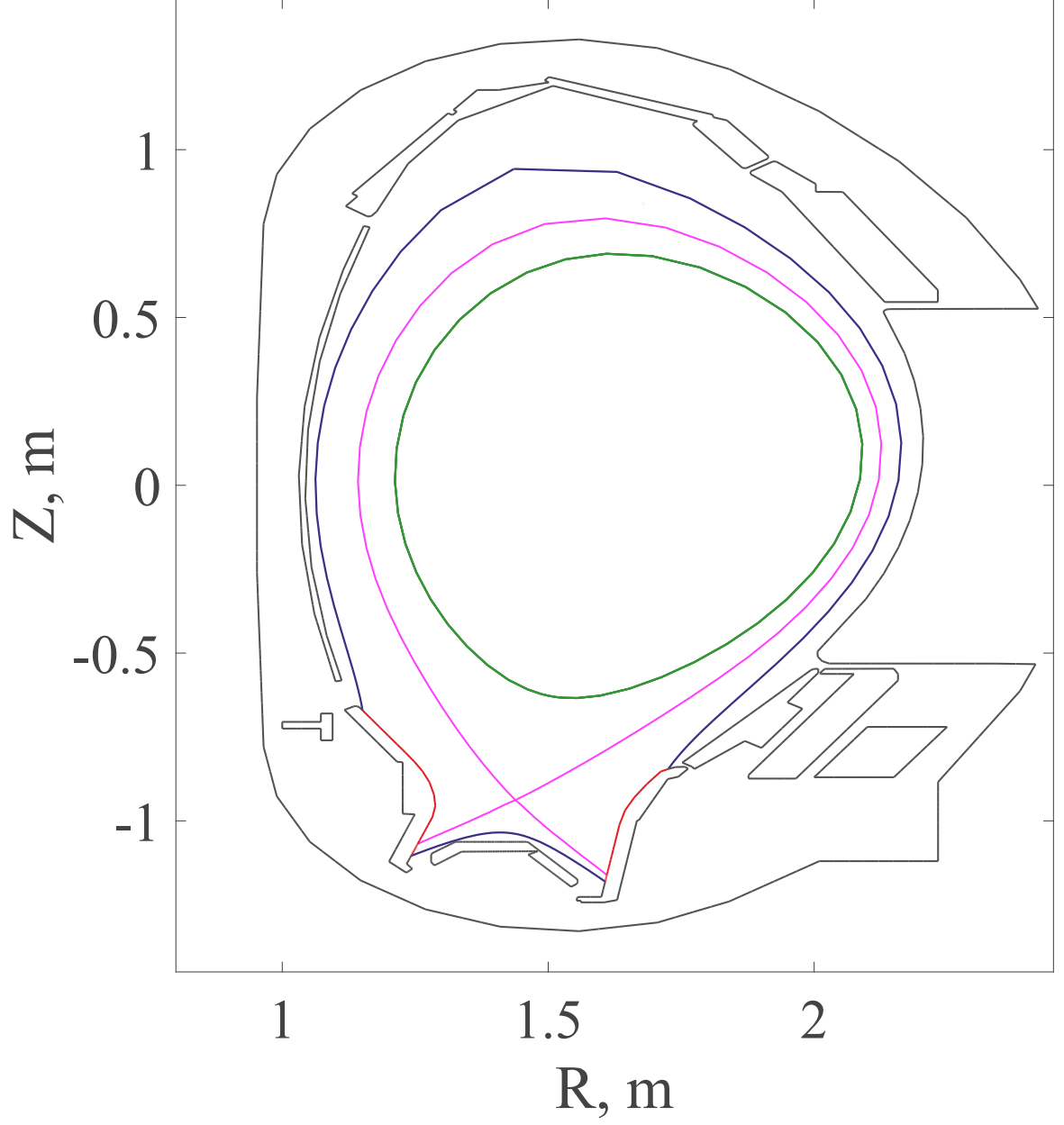


Figure 1: Surfaces used in particle balance. Green line: core boundary. Blue lines: wall boundary of B2 domain, which consists of northern boundary and southern PFR boundary. Red lines: target, i.e. the intersection of B2 domain and divertor plates. Grey lines: vacuum vessel elements, except targets. Magenta line: separatrix

1.2 Neutral flux through the core boundary, `ntr_core`

This is simply the integral of the neutral flux density over the core boundary, computed with the help of standard Eirene surface averaged tallies `PRF[A,M,I,PH,P][AT,ML]` and `POT[AT,ML]` according to the accepted sign convention.

Expected to be negative.

1.3 Ion flux to targets, `ion_targ`

Here and below target means the intersection of real divertor plates with the B2 computational domain. The rest of the plates are considered as “side walls”, since ions exist in the B2 domain only and cannot reach these parts of plate surface.

This is the net ion flux on targets. Summing over the all two or four targets and over all ionization states corresponding to given isonuclear sequence is assumed. The integral is computed basing on the `fna` array, since `fna_mdf` array has nothing with the ion flux.

Expected to be negative, since it is the ion sink.

1.4 Neutral flux to targets, `ntr_targ_tot`

This is simply the integral of the neutral flux density over target boundaries, computed with the help of standard Eirene surface averaged tallies `PRF[A,M,I,PH,P][AT,ML]` and `POT[AT,ML]` according to the accepted sign convention.

The flux consists of the reflected neutrals plus re-emitted neutrals minus impinging neutrals,

$$\text{PRFPAT} + \text{PRF}[A,M,I]AT - \text{POTAT} + \{\text{similar contribution from molecules}\}$$

in terms of Eirene tallies.

The first flux is assumed to be originating from the ions impinging onto the target surface.

If there is no sputtering or deposition for the species considered, the `ntr_targ_tot` is expected to be positive and equal to the absolute value of `ion_targ`.

1.5 Neutral flux through targets caused by ion recycling, `ntr_targ_rec`

This is the integral of the recycling neutral flux, which is assumed to be originating from the ions impinging onto the target surface.

Expected to be positive.

1.6 Neutral flux from targets due to sputtering, `ntr_targ_spt`

This is the neutral flux due to sputtering from targets. It is computed inside Eirene and is expected to be positive.

1.7 Pumping on targets, `ntr_targ_pmp`

This is a pumped flux computed in Eirene (tally `SPUMP`, transferred to B2.5 part through the array `wlpump`).

Expected to be zero for fully recycling species (like e.g. neon) or to be negative for those species which are allowed to be deposited on target (e.g. beryllium)

1.8 Ion flux to side wall, `ion_wall`

Here side wall means the northern boundary and southern Private Flux Region boundary of the B2 domain — the last boundary (in radial direction) accessible by ions in the model. The quantity `ion_wall` is just the integral of ion flux density (array `fna` independently of the running mode and on the value of the `b2tfnb_mdf` switch) over this boundary, again the summing over all ionization states corresponding to given isonuclear sequence is assumed.

Expected to be negative.

1.9 Neutral flux to side wall, `ntr_wall_tot`

For Eirene neutrals the side wall means the whole vacuum vessel boundary except targets, where targets is again the part of divertor plates which intersects the B2 domain. Naturally, this boundary is not accessible by ions in the model. The quantity `ntr_wall_tot` is the flux of recycling neutrals plus flux of re-emitted neutrals minus flux of impinging neutrals integrated over this surface. In terms of Eirene tallies it is again

$$\text{PRFPAT} + \text{PRF}[A,M,I]\text{AT} - \text{POTAT} + \{\text{similar contribution from molecules}\}.$$

Naturally, the first term (recycling neutrals), which are assumed to originate from impinging ion recycling, is zero.

For those species which are not sputtered or deposited this integrated flux is expected to be negative and equal to the pumping flux, since some side wall segments are absorbing surfaces thus representing a pump.

1.10 Neutral source due to ion recycling, `ntr_wall_rec`

This is the recycling neutrals flux integrated over the side walls as it is seen for neutrals (see the description of `ntr_wall_tot`) and for ions (see the description of `ion_wall`). The computation is based on the Eirene tallies `PRFPAT` and `PRFPML`, the accepted sign convention is used.

As the contribution from vacuum vessel is zero, and as the recycling (turning of ions into neutrals) takes place at the far SOL and far PFR boundaries of the B2.5 computational domain, the quantity `ntr_wall_rec` is expected to be positive and equal to the absolute value of the `ion_wall`. The `ntr_wall_rec` thus does not give a contribution to the `ntr_wall_tot`.

1.11 Neutral source due to sputtering, `ntr_wall_spt`

This is the neutral flux due to sputtering from side walls (i.e. from vacuum vessel elements). It is computed inside Eirene and is expected to be positive. Note that if there is effective sputtering source on the far SOL and PFR boundaries of B2.5 domain due to impinging ions, it is also counted here.

1.12 Pumping on side wall, `ntr_wall_pmp`

This is a pumped flux computed in Eirene (tally `SPUMP`, transferred to B2.5 part through the array `wlpump`). This is done exactly in the same manner as for quantity `ntr_wall_trg`

Should be equal to `ntr_wall_tot` exactly for fully recycling species, like e.g. neon.

1.13 Ion source due to ionization of neutral state, `src_ioniz`

This is the volume integral of the ionization source computed by Eirene for given ionuclear sequence. Expected to be positive.

Thus the ion balance (in the stationary state) should satisfy

$$\text{ion_core} + \text{ion_targ} + \text{ion_wall} + \text{src_ioniz} = 0$$

1.14 Neutral puffing/seeding rate, `ntr_puff`

This is just a positive constant equal to a prescribed puffing/seeding rate (variable `userfluxparm` for the stratum labeled by 'C').

1.15 Total flux into the whole domain, `flux_tot`

This is the sum of ion and neutral fluxes through the core boundary, ion and neutral fluxes through targets, neutral flux through the vacuum vessel boundary, neutral puffing/seeding rate, ion flux to the wall boundary of B2 domain (including the PFR) and the neutral source due to ion recycling on this boundary:

$$\begin{aligned} \text{flux_tot} = & \text{ion_core} + \text{ntr_core} + \text{ion_targ} + \text{ntr_targ_tot} + \\ & + \text{ion_wall} + \text{ntr_wall_tot} + \text{ntr_wall_rec} + \text{ntr_puff} \end{aligned}$$

In the stationary state it is expected to be zero.

1.16 Time derivative of the ion content in the whole domain, `ion_dn/dt`

Time derivative of the ion content in the whole domain, the summing over all ionization states corresponding to given isonuclear sequence is assumed.

2 Standalone mode

2.1 Ion flux through the core boundary, `ion_core`

Exactly the same as in coupled mode.

2.2 Neutral flux through the core boundary, `ntr_core`

This is a neutral flux through the code boundary computed basing on the array `fna`. Expected to be negative.

2.3 Ion flux to targets, `ion_targ`

Exactly the same as in coupled mode.

2.4 Neutral flux to targets, `ntr_targ_tot`

This is simply the integral of the neutral flux density over target boundaries, computed basing on the array `fna`.

If there is no sputtering or deposition for the species considered, the `ntr_targ_tot` is expected to be positive and equal to the absolute value of `ion_targ`.

2.5 Neutral flux through targets caused by ion recycling, `ntr_targ_rec`

This is the integral of the recycling or reflecting neutral flux, which is assumed to be originating from the ions impinging onto the target surface. Computed in the `b2stbr_phys.F`, (the array `sna_stbr_neut`, please see this routine for more details).

Expected to be positive.

2.6 Neutral flux from targets due to sputtering, `ntr_targ_spt`

This is the neutral flux due to sputtering from targets. Neutral sources due to thermal evaporation and radiation enhanced sublimation are added here as well. See the `b2stbr_phys.F` for details (the array `sna_stbr_sput`).

2.7 Pumping on targets, `ntr_targ_pmp`

This is a neutral flux prescribed by boundary conditions for neutrals (it is computed inside the `b2stbc.F`).

Depends on the boundary condition for neutrals on target. For example, if `BCCON=5` or `BCCON=8` are set with `CONPAR=0.0`, then the `ntr_targ_pmp` should be zero.

2.8 Ion flux to side wall, `ion_wall`

Exactly the same as in coupled mode.

2.9 Neutral flux to side wall, `ntr_wall_tot`

Here side wall means the northern boundary and southern Private Flux Region boundary of the B2 domain — the last boundary (in radial direction) accessible by ions in the model. The `ntr_wall_tot` is an integral of neutral flux (computed basing on the `fna` array) over this boundary.

2.10 Neutral source due to ion recycling on side wall, `ntr_wall_rec`

This is the integral of the recycling or reflecting neutral flux, which is assumed to be originating from the ions impinging onto the wall surface. Computed in the `b2stbr_phys.F`, (the array `sna_stbr_neut`, please see this routine for more details.)

Expected to be positive.

2.11 Neutral source due to sputtering on side wall, `ntr_wall_spt`

This is the neutral flux due to sputtering from targets. Neutral sources due to thermal evaporation and radiation enhanced sublimation are added here as well. See the `b2stbr_phys.F` for details (the array `sna_stbr_sput`).

2.12 Pumping on side wall, `ntr_wall_pmp`

This is a neutral flux prescribed by boundary conditions for neutrals. It is implicitly assumed that these boundary conditions are leakage ones, or boundary conditions which prescribe a decay length — i.e. the boundary conditions which assume the presence of neutral flux directed outside of the computational domain.

THOSE BOUNDARIES WHERE BOUNDARY CONDITIONS OF PRESCRIBED FLUX DENSITY (`BCCON=5`) OR PRESCRIBED TOTAL FLUX (`BCCON=8`) ARE APPLIED WITH THE FLUX DIRECTED INSIDE THE COMPUTATIONAL DOMAIN (`CONPAR>0`) ARE TREATED AS BOUNDARIES OF PUFFING. SUCH BOUNDARIES DO NOT GIVE CONTRIBUTION TO `ntr_wall_pmp`.

2.13 Ion source due to ionization of neutral state, `src_ioniz`

This is the volume integral of the ionization source from neutral state for given isonuclear sequence. Expected to be positive.

Thus the ion balance (in the stationary state) should satisfy

$$\text{ion_core} + \text{ion_targ} + \text{ion_wall} + \text{src_ioniz} = 0$$

2.14 Neutral puffing/seeding rate, `ntr_puff`

This is a neutral flux prescribed by boundary conditions for neutrals on those boundaries which are treated as puffing boundaries (i.e. where `(BCCON=5)` or `(BCCON=8)` with `(CONPAR>0)`, see the description above.

2.15 Total flux into the whole domain, `flux_tot`

Exactly the same as in coupled mode.

2.16 Time derivative of the ion content in the whole domain, `ion_dn/dt`

Exactly the same as in coupled mode.